LF0271-003-D01-00

89 mm

MUCERA syrup

15 mg/5 mL

Composition:
Each 5 mL syrup contains:
Ambroxol HCl ..... 15 mg

Preservative:
Sodium Benzoate ..... 12.5 mg

Product Description:
Red solution, sweet and slightly bitter taste with strawberry odor.

Pharmacodynamics:
The major pharmacodynamic actions of ambroxol are surfactant stimulation, mucokinetic and secretagogue activity. The therapeutic activities of the drug in animal models of the infant and adult respiratory distress syndrome (IRDS and ARDS) are reviewed. SO2 exposed rats, which exhibited increased airway resistance and work of breathing, were used as an animal model of a bronchitic syndrome. The active group was treated with 25 mg kg-1 oral ambroxol for 10 days. The airway resistance of this group (53.6 +/- 7.0 Pa.mL-1.s) was significantly lower than that of the control (81.2 +/- 11.4). Specific work of breathing was also lower in the treated group (0.26 +/- 0.02 mJ.mL-1, control: 0.35 +/- 0.029). The alleviation of airflow limitation was a consequence of subacute treatment and not of acute bronchodilatation. Treatment with the beta 2-adrenergic drug clenbuterol further improved both active and placebo groups.
The mucokinetic activity of ambroxol was demonstrated as an increase in mucus or particle transport rate on the frog palate and in guinea pig tracheas. To clarify the mechanism we examined the influence of the drug on the ciliary beat frequency (CBF) of isolated cells from guinea pig tracheas. Ambroxol significantly stimulated the CBF up to 10.8%, salbutamol up to 19.5%. The effect of ambroxol on CBF certainly contributes to, but does not explain the more pronounced increase in transport rate. Thus a stimulation of the mucociliary transport rate by the other major pharmacodynamic mechanisms, secretagogue activity and stimulation of pulmonary surfactant, has to be considered. The therapeutic activity of ambroxol in models of the infant and adult respiratory distress syndrome and of the chronic bronchitic syndrome is reviewed.

Pharmacokinetics:
Pharmacokinetics and biotransformation of trans-4-(2-amino-3,5-dibromo-benzylamino) cyclohexanol hydrochloride (ambroxol, NA 872 Cl) was studied using the 14C-labelled compound. Absorption after oral administration was found to be fast and complete. Elimination half-life of radioactivity in the blood was estimated as 20-25 hours in rat, dog and man and as 2 hours only in rabbit. This apparent elimination half-life is governed by the disposition of acidic metabolites of NA 872. In man and rabbit radioactivity is excreted almost completely into the urine, whereas in rat and dog biliary excretion is also observed. Routes of biotransformation are similar in all 4 species. NA 872 is metabolized by phase I reactions to NA 873 and finally to dibromoanthranilic acid. Phase II reactions with the parent compound and metabolites are observed mainly in man and rabbit.
In humans, ambroxol is metabolized to dibromoanthranilic acid (DBAA) and 6,8-dibromo-3-(trans-4-hydroxycyclohexyl)-1,2,3,4-tetrahydroquinazoline (DHTQ). The formation of DHTQ proceeds non-enzymatically, whereas that of DBAA requires NADPH. Studies have been performed to identify the CYP isozyme(s) involved in the formation of DBAA using human liver microsomes and microsomes expressing recombinant human CYP isozymes (1A1, 1A2, 2A6, 2B6, 2C8, 2C9, 2C19, 2D6, 2E1, 3A4 and 4A11). The apparent Vmax and Km for the formation of DBAA were 472±192 pmol/ min/mg protein and 248 ± 40.6 μM respectively (mean ± S.D., n = 3). Of the recombinant CYP examined, only CYP3A4 metabolized ambroxol to DBAA. The apparent Vmax and Km were 1.42 pmol/min/pmol P450 and 287 μM respectively. Among the CYP inhibitors examined (furafylline, sulphaphenazole, quinidine, diethyldithiocarbamic acid, ketoconazole), only ketoconazole inhibited the production of DBAA (> 80%) at 1 μM and anti-CYP3A antiserum almost completely inhibited the formation of DBAA. These results suggest that CYP3A4 is predominantly involved in the metabolism of ambroxol to DBAA in humans.

Indication:
MUCERA Syrup is indicated as secretolytic therapy in acute & chronic bronchopulmonary diseases associated with abnormal mucus secretion & impaired mucus transport.

Contraindication:
MUCERA Syrup should not be used in patients known to be hypersensitive to ambroxol or other components of the formulation.

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Warning and Precautions:
It is not recommended for long treatment without medical supervision.
In a few case of renal failure, metabolite of ambroxol will be accumulated in liver.
Patients with rare hereditary fructose intolerance should not take this medicine.
Very rare cases of chronically associated severe skin impairments such as Stevens Johnson Syndrome, Toxic Epidermal Necrolysis (TEN), Erythema Multiforme (EM) and Acute Generalized Exanthematous Pustulosis (AGEP) have been reported. In most cases, these could be explained by the severity of the underlying disease or concomitant administration of another drug. In the early stages of such severe skin reactions, initially only nonspecific flu-like symptoms appear, e.g. fever, arthralgia, runny nose, cough, and sore throat. If skin or mucous membrane damage occurs, seek medical advice immediately and discontinue treatment as a precaution.

Interaction with Other Medicaments:
Administration of ambroxol HCl together with antibiotics (Amoxycillin, Cefuroxime, Erythromycin, Doxycycline) leads to higher antibiotic concentration in the lung tissue. No clinically relevant unfavourable interaction with other medications have been reported.

Pregnancy and Lactation:
- Ambroxol crosses the placental barrier. Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/foetal development, parturition or postnatal development. Extensive clinical experience after the 28th week of pregnancy has shown no evidence of harmful effects on the foetus. Nonetheless, the usual precautions regarding the use of drugs during pregnancy should be observed. Especially during the first trimester, the use of MUCERA Syrup is not recommended. The benefit of the drug in pregnant women during first trimester should be weighed against a possible risk to the fetus.
- Ambroxol is excreted in breast milk. Therefore, MUCERA Syrup is not recommended for use in nursing mothers. However unfavourable effects on breastfed infants would not be expected.

Side Effects:
- Occasional, mild side effects from MUCERA Syrup include: Nausea, diarrhea, dyspepsia (e.g. indigestion, bloated feeling), vomitting, abdominal pain, heartburn, a feeling of fullness in the stomach (a bloated feeling), dysgeusia (e.g. changed taste), oral and pharyngeal hypoaesthesia, dry mouth, dry throat.
- Symptoms of allergic reactions may include: Skin rashes, difficulty in breathing, swelling of the face, lips, mouth, tongue or throat which may cause difficulty swallowing or breathing.
- Not known effect on the ability to drive or run the machine.
- Immune System Disorders.
  Frequency not known: Anaphylactic reactions including anaphylactic shock
- Skin and Subcutaneous Skin Disorders
  Frequency not known: Severe skin reactions (including Stevens Johnson syndrome, Toxic epidermal necrolysis (TEN), Erythema Multiforme (EM) and Acute Generalized Exanthematous Pustulosis (AGEP).

Recommended Dose:
Children up to 2 years : 2.5 mL of syrup, 2 times daily.
Children 2-5 years : 2.5 mL of syrup, 2-3 times daily.
Children 5-12 years : 5 mL of syrup, 2-3 times daily.
Children's dose may also be calculated based on body weight: 1.5 to 2 mg/kg body weight/day in 2 divided doses. To be taken after food.

Route of Administration:
Oral route.

Symptoms and Treatment of Overdose:
No symptoms of overdosage have been reported in man to date. If they occur, symptomatic treatment should be provide.

Storage Condition:
Store below 30°C. Protect from light.
Keep medicine out of reach of children.

Shelf Life:
2 years from manufacturing date.

Presentations:
Bottle of 60 mL syrup.

Date of revision : AUG 2024

Manufactured by:
PT Otto Pharmaceutical Industries
Jl. Dr. Setiabudhi Km 12.1
Bandung 40391 – Indonesia

Product Registration Holder:
The Zyfas Medical Co.
No. 15, Jalan Mega 1/5
Taman Perindustrian Nusa Cemerlang,
79200 Iskandar Puteri, Johor, Malaysia.

LF0271-003

200 mm

bagian belakang

Item	**LEAFLET MUCERA SYRUP EXPORT MALAYSIA**
Kode	LF0271-003
Ukuran	200 x 89 mm (2 muka)
Bahan	HVS 60 gsm
Warna	Hitam
Type & besar huruf	**Brand :** Charter BdBT (20 pt) **Lain-lain :** Arial, SwitzerlandCondensed
Barmark	Terletak di sebelah kiri leaflet, 8 cm dari atas, ukuran 3 x 3 mm (3 buah) jarak 2 mm, warna Hitam
Catatan	Tambahkan kode supplier di sebelah kanan kode kami
Catatan Perubahan	Revisi redaksi

Print out ini hanya menunjukkan komposisi warna bukan acuan cetak

Lipatan akhir